

Sameeran Joshi

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EDUCATION

Ph.D. in Computer Science

Aug 2022 – Present

University of Utah, Salt Lake City, UT

Advisor: Dr. Mary Hall. Graduate Research Assistant, CTOP Group. Research focus: compilers and programming models for spatial dataflow hardware, hardware–software co-design, and high-performance computing.

B.E. in Computer Science

May 2015 – Apr 2019

P.V.G.'s College of Engineering and Technology (COET), Pune, India

RESEARCH INTERESTS

Programming languages, compilers, and high-performance computing, with a focus on designing new abstractions and compilation techniques for existing and emerging spatial dataflow architectures.

PUBLICATIONS

Journal Articles

- M. Hall, C. Oancea, A. C. Elster, A. Rasch, **S. Joshi**, A. M. Tavakkoli, R. Schulze. “Scheduling Language Chronology: Past, Present, and Future.” *ACM Transactions on Architecture and Code Optimization (TACO)*, 2025.

Conference and Workshop Papers

- S. Joshi**, O. Antepara, S. Williams, H. Johansen, M. Hall. “GLOW: Accelerating Geometric Multigrid Levels on Cerebras Wafer-Scale Engine.” *Proceedings of the International Conference for High Performance Computing, Networking, Storage, and Analysis (SC26)*, to appear.
- A. M. Tavakkoli*, **S. Joshi***, S. Singh, Y. Xu, P. Sadayappan, M. Hall. “PEAK: Generating High-Performance Schedules in MLIR.” *36th International Workshop on Languages and Compilers for Parallel Computing (LCPC)*, Oct. 2023. (*equal contribution)
- M. Hall, G. Gopalakrishnan, E. Eide, J. Cohoon, J. M. Phillips, M. Zhang, S. Y. Elhabian, A. Bhaskara, H. Dam, A. Yadrov, T. Kataria, A. M. Tavakkoli, **Sameeran Joshi**, M. S. T. Karanam. “An NSF REU Site Based on Trust and Reproducibility of Intelligent Computation: Experience Report.” *EduHPC Workshop*, held at SC23.

RESEARCH AND INDUSTRY EXPERIENCE

Compiler Intern, CSL Compiler Team

May 2026 – Aug 2026

Cerebras Systems

Manager: Salem Derisavi. Worked on the CSL compiler team; built a non-actor-based programming model in MLIR to automatically generate CSL code.

Compiler Intern, AIE Accelerator (NPU)

May 2025 – Aug 2025

AMD, Campbell, CA

Manager: Mahesh Ravishankar. Worked on vectorization support in the AMD NPU and IREE compiler stack.

Research Intern, Technical PhD

May 2024 – Aug 2024

Argonne National Laboratory, Lemont, IL

Supervisor: Dr. Siddhisanket Raskar. Worked on ALCF AI testbed hardware and the GraphCore backend for DaCe.

CPU Compiler Engineer, Full Time

Jun 2019 – Jun 2022

AMD, Bangalore, India

Worked on Flang, performance analysis tooling (internal conference paper), and verification of LLVM.

Undergraduate Researcher, GCC Project

Jun 2018 – Apr 2019

Remote

Extended Csmith with GNU C language extensions and found 12 critical bugs in the GCC upstream compiler.

HONORS AND AWARDS

- Student Travel Grant, Workshop on Sparse Tensor Computations
- Student Travel Grant, SC25 and SC26 (Volunteer)
- Tuition Fee Waiver Scheme (TFWS) Scholarship, undergraduate studies

PROFESSIONAL SERVICE

- Artifact Evaluation Committee: CGO 2025, ASPLOS 2025
- Program Committee Member: LLVM Developers’ Meeting 2021
- Volunteer: CppCon 2021, CppOnSea 2021